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Aragya Goyal
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EDUCATION

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|---|---------------------------------|
| University of Pittsburgh (Swanson School of Engineering/Frederick Honors Col.) | Pittsburgh, PA |
| • <i>B.S. - Computer Engineering (Autonomous Systems Focus); GPA: 3.99/4.00</i> | <i>August 2022 - April 2026</i> |
| Courses: Data Structures and Algorithms, Embedded Processors, Microelectronics, Digital Circuits | |

PROFESSIONAL EXPERIENCE

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| Carnegie Mellon University Robotics Institute | Pittsburgh, PA |
| • <i>Undergraduate Robotacist Intern</i> | <i>April 2023 - Present</i> |
- **ZOË 2 Rover:** (tinyurl.com/zoe2rover)
 - * Developing low-level software stack for the 2nd generation Zoë Rover set to conduct science in the Atacama Desert including CANopen protocols via the ros.canopen package for ROS2 to communicate with encoders and motors.
 - * Conducted motor and motor controller datasheet specifications validation via physical testing.
 - **Underwater Snake Robot:** (tinyurl.com/humrsCMU)
 - * Implemented High-Frequency Injection methods in Brushless DC thrusters to achieve control at low/zero speeds thus reducing minimum speed by 80% allowing for reduced oscillations in station-keeping control loops.
 - * Deployed station-keeping feature using AprilTags, IMU readings, and Nested PID Controllers to perform robot state-estimation underwater.
 - * Conducted major repairs on the buoyancy module of the robot and assisted in continual electrical maintenance.
 - **Apple's E-Waste Recycling Project:** (tinyurl.com/applecmu)
 - * Created datasets consisting of thousands of images for Machine Learning Models to detect screws in e-waste images.
 - * Integrated ROS1 and Python packages to track AprilTags using a Realsense camera for localization of UR3 robotic arm.
 - * Manufactured custom AprilTags using lasercutters and sheet metal manufacturing methods.

STUDENT ORGANIZATIONS

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| University Rover Challenge (Robotics and Automation Society) | Pittsburgh, PA |
| • <i>Chief Engineer/Integration Lead</i> | <i>August 2022 - Present</i> |
- Leading a multidisciplinary team of 30 students in engineering a rover, coordinating integration efforts across mechanical, electrical, software, and science teams. (tinyurl.com/roverimages)
 - Proposed and spearheaded the development of a prototype robotic hand utilizing pneumatics. (tinyurl.com/hydraarm)
 - Securing and managing over \$7,000 in funding to support team development and future growth.
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| FIRST and VEX Robotics | Exton & Royersford, PA |
| • <i>Team Captain/Design Engineer</i> | <i>August 2018 - June 2022</i> |
- Mentored younger students in Solidworks and manufacturing through workshops, enabling three-time Worlds qualification.
 - Won the VEX Judges Award, FIRST Excellence in Engineering Award, FIRST Industrial Design Award, and the FIRST Chairman's Award. (VEX Link: bit.ly/3w7a6b1) (FIRST Link: tinyurl.com/dwbot17)

PROJECTS

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- **555 Timer Roulette Board:**
 - **Bop It Project:**
 - **Micromouse Project:**
 - **Trains Project:**
 - **An Artistic take on Bipedal Robots:**
 - **STM-32 Elevator Simulator (ARM Assembly, Project Integration):** Designed and Implemented software architecture in ARM Assembly to operate a physical elevator simulator PCB. (tinyurl.com/stmelevator)
 - **Custom Cane (Human Centered Design, Solidworks):** Designed and fabricated a Walker-Cane Fusion to increase bathrooms accessibility for wheelchair users. Won first place at the Senior Design Expo. (tinyurl.com/CustomCane)
 - **Formula SAE E-Brake Bias (Solidworks):** Developed an e-brake bias system for a formula style racecar, utilizing Solidworks and 3D printing technology to enhance performance and usability. Won the FSAE Innovation Award for the design and implementation of the project. (tinyurl.com/ebreakb)
 - **Formula SAE Low-Cost Slip Angle Sensor (OpenCV, Raspberry Pi):** Worked to design and code prototypes of sensors which would allow for validation of slip angle using mouse sensors, digital cameras, and IMU's.
 - **String Art Generator and Optimizer (Research, Python):** Developed an innovative string art optimization tools and GUI using Python programming to improve upon existing string art generators. (tinyurl.com/goyalstring)

SKILLS AND AWARDS

- **Languages:** Python, C++, C, ARM Assembly, RISC-V Assembly
- **Technologies:** Linux, ROS1, ROS2, Docker, Github, Solidworks, KiCAD, MATLAB, Microsoft Suite, Microcontrollers, OpenCV
- **Manufacturing:** Milling, Soldering, Laser Cutting, General Shop Tools
- **Awards:** Dean's Honor List (2021-Present), Honor List (2021-Present), Eagle Scout

PUBLICATIONS

- M. Nye, A. Raji, A. Saba, E. Erlich, R. Exley, A. Goyal, A. Matros, R. Misra, M. Sivaprakasam, M. Bertogna, D. Ramanan, and S. Scherer, “**BETTY Dataset: A Multi-modal Dataset for Full-Stack Autonomy,**” in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 2025. [Online]. Available: (arxiv.org/abs/2505.07266)